

Species-Aware Module Review: dTDP-L-rhamnose Biosynthesis in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)
Commissioned bucket: KEGG ppu00523 "Polyketide sugar unit biosynthesis" (module area: other_kegg_pathway) **True module:** dTDP-L-rhamnose biosynthesis (soluble nucleotide-sugar supply, α -D-glucose-1-phosphate $\rightarrow$ dTDP-L-rhamnose) **Evidence tier throughout:** All KT2440 genes are UniProt Protein-Existence level **PE3 "Inferred from homology"**; no direct KT2440 enzymology exists. Functional transfer rests on strong homology to experimentally characterized *P. aeruginosa* and generic bacterial Rml enzymes.

1. Executive summary

The dTDP-L-rhamnose module is **fully satisfiable** in *P. putida* KT2440. All four canonical enzymatic steps are encoded by a single **contiguous *rmlBDAC*-type operon, PP_1785–PP_1782** (minus strand): `rffG(rmlB) – rfbD(rmlD) – rfbA(rmlA) – rfbC(rmlC)`. Each gene carries the correct family-specific InterPro/Pfam signature and the correct EC, and the operon layout is syntenic with the experimentally validated *P. aeruginosa* *rmlBDAC* cluster (

P 11065359

Two of the six candidate genes are **operon-external paralogs** that inflate the apparent step count: a second epimerase **PP_0265 (*rmlC*)** and a second reductase **PP_0500**. PP_0265 is a domain-confirmed, close (66.5% id) genuine RmlC. PP_0500 is a **divergent (39.7% id), genomically isolated** RmlD-family member (GenBank "...reductase-related protein") whose *physiological* substrate/role is unresolved.

The commissioned bucket **ppu00523 is a KEGG multi-mapping artifact**, not a real KT2440 polyketide pathway; the same shared nucleotide-sugar enzymes also map to ppu00521/00525/00520/00541. The generic module boundary (four soluble steps; LPS/O-antigen as consumers) is **correct for KT2440** and needs no revision.

Bottom line for curation: mark all four steps `covered` on the operon set; annotate PP_0265 as a confirmed redundant paralog; mark PP_0500 `candidate_uncertain` and promote it to full `fetch-gene` review.

2. Target-organism pathway definition

Process included (soluble module): the four-reaction conversion of α -D-glucose-1-phosphate into the activated deoxy-sugar donor dTDP- β -L-rhamnose:

1. **dTDP-glucose formation** — glucose-1-phosphate thymidyltransferase, EC 2.7.7.24 (dTTP + α -D-glucose-1-P $\rightarrow$ dTDP- α -D-glucose + PPi).
2. **4,6-dehydration** — dTDP-glucose 4,6-dehydratase, EC 4.2.1.46 ($\rightarrow$ dTDP-4-dehydro-6-deoxy- α -D-glucose; NAD⁺-dependent, internal redox).
3. **3,5-epimerization** — dTDP-4-dehydrorhamnose 3,5-epimerase, EC 5.1.3.13 ($\rightarrow$ dTDP-4-dehydro- β -L-rhamnose).
4. **Final reduction** — dTDP-4-dehydrorhamnose reductase, EC 1.1.1.133 (NADPH-dependent $\rightarrow$ dTDP- β -L-rhamnose).

Neighboring processes to keep separate (consumers, NOT part of the module): downstream L-rhamnosyl transfer into LPS core/O-antigen and other glycans. In *Pseudomonas*, L-rhamnose is a documented component of the LPS core (the attachment receptor for A-band/B-band O antigen) and of O-antigen polysaccharides (*P. aeruginosa* evidence;).

P 11065359

Note a species caveat: rhamnolipid surfactant biosynthesis, a well-known L-rhamnose sink in *P. aeruginosa*, is **not** an established feature of *P. putida* KT2440 — do not assume it as a sink here.

Alternate names / database definitions: - Gene symbols: *rmlA/rfbA* (=rffH variants), *rmlB/rffG*, *rmlC/rfbC*, *rmlD/rfbD*. KT2440 uses mixed *rfb/rff/rml* symbols. - The pathway is captured as MetaCyc "dTDP-L-rhamnose biosynthesis I" and UniProt "Carbohydrate biosynthesis; dTDP-L-rhamnose biosynthesis." - KEGG has **no dedicated dTDP-L-rhamnose map**; the enzymes appear inside ppu00521 (Streptomycin), ppu00523 (Polyketide sugar unit), ppu00525 (Acarbose/validamycin), ppu00520 (Amino/nucleotide sugar metabolism), ppu00541, and global maps 01100/01110/01250. **ppu00523 should be treated as a label, not the pathway.**

3. Expected step model (with target-taxon status)

Step	Reaction (EC)	Expected gene family	KT2440 gene(s)	Status
1	Glc-1-P thymidylylation (2.7.7.24)	RmlA / G1P- thymidylyltransferase	PP_1783 / rfbA	covered
2	dTDP-glucose 4,6- dehydration (4.2.1.46)	RmlB / dTDP-glucose dehydratase	PP_1785 / rffG	covered
3	3,5-epimerization (5.1.3.13)	RmlC epimerase	PP_1782 / rfbC (+ paralog PP_0265 / rmlC)	covered (one redundant paralog)
4	NADPH reduction (1.1.1.133)	RmlD reductase	PP_1784 / rfbD (+ paralog PP_0500)	covered (one uncertain paralog)

No step is missing; no step is "not expected." The pathway is complete on the single operon PP_1782–PP_1785.

4. Candidate genes and evidence

Operon PP_1782–PP_1785 (minus strand; syntenic *rmlBDAC*):

- **PP_1783 / rfbA (Q88LZ3) — glucose-1-P thymidylyltransferase, EC 2.7.7.24 (Step 1).** InterPro IPR005907 (G1P_thy_trans_s), Pfam PF00483, PANTHER PTHR43532:SF1; Mg²⁺/nucleotidyltransferase keywords. UniProt catalytic-activity and dual pathway annotation (LPS O-antigen; dTDP-L-rhamnose). Evidence: homology (PE3). **Confident RmlA.** Caveat: KEGG "primary bucket" ppu00525 is a multi-mapping artifact — it is a core member of the rhamnose module.
- **PP_1785 / rffG (Q88LZ1) — dTDP-glucose 4,6-dehydratase, EC 4.2.1.46 (Step 2).** InterPro IPR005888 (dTDP_Gluc_deHydtase), Pfam PF16363, NAD-binding. Named "dTDP-glucose 4,6-dehydratase 2" in GenBank. Evidence: homology (PE3). **Confident RmlB.** Same ppu00525 multi-map caveat.

- **PP_1782 / rfbC (Q88LZ4) — dTDP-4-dehydrorhamnose 3,5-epimerase, EC 5.1.3.13 (Step 3).** InterPro IPR000888 (RmlC-like cupin), Pfam PF00908, PANTHER PTHR21047:SF2. Evidence: homology (PE3). **Confident RmlC**, and the operon-resident (physiologically committed) epimerase.
- **PP_1784 / rfbD (Q88LZ2) — dTDP-4-dehydrorhamnose reductase, EC 1.1.1.133 (Step 4).** InterPro IPR005913 (dTDP_dehydrorham_reduct) + IPR029903 (RmlD-like-bd), Pfam PF04321, NADP. Evidence: homology (PE3). **Confident RmlD**, operon-resident committed reductase.

Operon-external paralogs:

- **PP_0265 / rmlC (Q88R69) — EC 5.1.3.13, ~319.8 kb.** Identical domain architecture to PP_1782 (IPR000888/PF00908/PTHR21047:SF2); **66.5% aa identity** to PP_1782. **A genuine second RmlC epimerase.** Genomic neighbors are unrelated (ADP-sugar pyrophosphorylase, siderophore receptor, sensor kinases) — not in a deoxy-sugar cluster; likely a redundant/spare epimerase or one channeled to another dTDP-4-keto-6-deoxy-hexose.
- **PP_0500 (Q88QJ2) — EC 1.1.1.133, ~589.9 kb.** Carries the diagnostic RmlD-family signature (IPR005913/PF04321), so the reductase family assignment is **domain-supported**; however it is only **39.7% identical** to the operon RmlD (PP_1784) — barely above background — has **no gene symbol**, GenBank name "dTDP-4-rhamnose reductase-related protein," UniProt annotation score 2.0, and sits beside a pseudogene (PP_0499) and a lone NAD-epimerase (PP_0501) with **no cognate RmlA/RmlB partner**. **Genuine RmlD-family, but physiological substrate/role uncertain.**

Broad/spurious mappings noted: Both reductases inherit a wrong PANTHER *subfamily* tag PTHR10491:SF4 "methionine adenosyltransferase 2 subunit beta" (a PANTHER SF mis-grouping artifact; the family PTHR10491 = dTDP-4-dehydrorhamnose reductase is correct). Do not propagate the SF4 label.

5. Gaps, ambiguities, and likely over-annotations

- **No true pathway gap.** All four steps are covered by the operon.
- **Paralog ambiguity (Step 3 and Step 4).** Two epimerases and two reductases. The operon copies (PP_1782, PP_1784) are the committed enzymes; PP_0265 and PP_0500 are extra copies. Curatorially, steps should not be double-counted.

- **PP_0500 is the key uncertainty.** Domain signature supports EC 1.1.1.133, but low identity + orphan location + "related protein" naming leave open whether it (a) is a redundant dTDP-L-rhamnose reductase, or (b) reduces a different dTDP-4-keto-6-deoxy-sugar. This is the most likely locus of over-precise EC transfer.
- **KEGG bucket over-interpretation.** ppu00523 "Polyketide sugar unit biosynthesis" does **not** imply KT2440 makes polyketide sugars, streptomycin, acarbose, or validamycin (ppu00521/00525) — these are shared-enzyme labels. Treat ppu00523 as a boundary artifact.
- **Evidence ceiling.** Every conclusion for KT2440 is homology-based (PE3). No target-strain knockout, complementation, or enzymology exists.

6. Module and GO-curation recommendations

Per-step module status: - Step 1 (EC 2.7.7.24) → **covered** by PP_1783/rfbA. - Step 2 (EC 4.2.1.46) → **covered** by PP_1785/rffG. - Step 3 (EC 5.1.3.13) → **covered** by PP_1782/rfbC; annotate **PP_0265** as a confirmed redundant paralog (secondary, not step-satisfying on its own). - Step 4 (EC 1.1.1.133) → **covered** by PP_1784/rfbD; annotate **PP_0500** as **candidate_uncertain** paralog.

Module boundaries: the generic four-step module (α -D-glucose-1-P → dTDP-L-rhamnose; LPS/O-antigen as consumers) is **correct for KT2440 — no revision needed** (`module_needs_revision = No`). Fold rfbA/rffG into this module regardless of their ppu00525 KEGG tag.

Bucket handling: reassign the working bucket from KEGG ppu00523 to the MetaCyc/UniProt "dTDP-L-rhamnose biosynthesis" definition, or explicitly annotate ppu00523 as a shared-enzyme label.

GO curation: existing GO/EC assignments (GO:0008830 dTDP-4-dehydrorhamnose 3,5-epimerase; GO:0008831 dTDP-4-dehydrorhamnose reductase; GO:0008879 glucose-1-phosphate thymidyltransferase; GO:0008460 dTDP-glucose 4,6-dehydratase) are appropriate for the operon genes. **No new GO term request is needed.** For PP_0500, prefer the family-level term with an IEA/ISS evidence code and a curator note on substrate uncertainty rather than an over-specific assertion.

7. Genes to promote to full fetch-gene review

1. **PP_0500 (highest priority)** — divergent, orphan RmlD-family paralog; resolve whether it is a redundant dTDP-L-rhamnose reductase or a reductase for a different dTDP-4-keto-6-deoxy-sugar. Check catalytic residues, cofactor (NADPH vs NADH), operonic context of neighbor PP_0501, and any phenotype data.
2. **PP_0265** — confirm second RmlC and clarify why KT2440 retains two epimerases (redundancy vs distinct sugar route); moderate priority.
3. (Optional) **PP_1782–PP_1785 operon** — a single consolidated fetch-gene review to attach the operon/syntenic evidence and mark the four steps covered.

8. Key references

- **Rahim R, Burrows LL, Monteiro MA, Perry MB, Lam JS.** Involvement of the *rml* locus in core oligosaccharide and O polysaccharide assembly in *Pseudomonas aeruginosa*. *Microbiology* 2000.

P 11065359

— Characterizes the *rmlBDAC* operon making dTDP-L-Rha; L-Rha is the LPS-core receptor for O-antigen attachment. *Species transfer to KT2440: strong (same genus, syntenic operon), but still homology-level for KT2440.*

- **Dong C, et al. (Naismith group).** RmlC epimerase mechanism via an unusual twist-boat intermediate. (2007).

P 17046787

— Mechanism/validated drug target for the EC 5.1.3.13 step. *Generic bacterial evidence.*

- **Christendat D, et al.** Crystal structure of dTDP-4-keto-6-deoxy-hexulose 3,5-epimerase (RmlC). (2000).

P 10827167

— Structural basis and conserved active site of RmlC. *Generic bacterial evidence.*

- **UniProt** entries Q88LZ3 (rfbA), Q88LZ1 (rffG), Q88LZ4 (rfbC), Q88LZ2 (rfbD), Q88R69 (rmlC/PP_0265), Q88QJ2 (PP_0500) — annotations, EC, InterPro/Pfam, PE3 evidence tier.

- **KEGG** ppu:PP_0265/PP_0500/PP_1782-1785 — genomic coordinates, operon synteny, and multi-map pathway memberships (ppu00521/00523/00525/00520/00541).
 - **InterPro/Pfam** families: IPR005907/PF00483 (RmlA), IPR005888/PF16363 (RmlB), IPR000888/PF00908 (RmlC), IPR005913/PF04321 (RmlD).
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Evidence provenance summary

- **Direct KT2440 experiment:** none identified.
 - **Homology/synteny (strong):** operon completeness and step assignments, transferred from *P. aeruginosa* rmlBDAC + universal Rml family signatures.
 - **Database/computational:** EC, InterPro/Pfam domains, genomic coordinates, pairwise identities (Needleman-Wunsch computed here).
 - **Open questions for experts:** (1) in-vivo substrate/role of PP_0500; (2) reason for retained duplicate epimerase PP_0265; (3) the physiological L-rhamnose sink(s) in KT2440 (LPS core/O-antigen assumed; rhamnolipid unconfirmed).
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